

RUKMANI KUMARI

GEOSPATIAL ANALYST | GIS & REMOTE SENSING | GEOGRAPHY

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PROFESSIONAL PROFILE

Research-driven Geography professional and PhD Scholar with practical expertise in GIS, Remote Sensing, spatial analysis, cartography, and environmental applications. Experienced in QGIS and ArcGIS-based LULC mapping, change detection, NDVI, Land Surface Temperature (LST), DEM and terrain analysis, thematic cartography, and satellite-data interpretation. Combines academic research, geospatial project work, and analytical skills to develop evidence-based spatial outputs for environmental and geographic applications.

CORE COMPETENCIES

- GIS & Spatial Analysis
- ArcGIS / ArcMap
- LULC Classification & Change Detection
- Land Surface Temperature (LST)
- Watershed & Drainage Analysis
- Raster & Vector Analysis
- Data Interpretation & Analysis
- Microsoft Office 365
- QGIS
- Remote Sensing & Satellite Analysis
- NDVI / NDWI / MNDWI / NDBI
- DEM & Terrain Analysis
- Georeferencing & Digitization
- Thematic Cartography
- Academic Research & Technical Writing

ACADEMIC CREDENTIALS

Doctor of Philosophy (PhD) in Geography *Ongoing*

SNDT Women's University, Pune, Maharashtra. Research focus: sustainable development, environmental geography, and spatial analysis.

Post Graduate Diploma in Geoinformatics (PGDGI) *July 2026 session*

Indira Gandhi National Open University (IGNOU). Geospatial training in GIS, Remote Sensing, spatial analysis, and geoinformatics applications.

Master of Arts (MA) in Geography *January 2022*

Pataliputra University, Patna, Bihar. University Gold Medallist; ranked 1st in a cohort of 200+ students; 75% (7.94 CGPA).

Bachelor of Arts Hons. in Geography *January 2017*

BRABU (Babasaheb Bhimrao Ambedkar Bihar University), Muzaffarpur. First Class Honours; 67% academic standing.

UGC NET QUALIFICATIONS

UGC NET — PhD Eligibility — Result declared: 22 February 2025

UGC NET — Assistant Professorship — Result declared: 24 July 2023

SELECTED GEOSPATIAL PROJECTS

LULC Change Analysis, 2010–2025

Prepared LULC maps and a 2010–2025 transformation map using GIS and Remote Sensing techniques; analyzed temporal land-use/land-cover change.

NDVI Analysis

Prepared satellite-derived NDVI outputs to assess vegetation conditions and spatial/temporal variation.

Land Surface Temperature (LST)

Prepared LST outputs for spatial temperature analysis and comparison with vegetation and urban characteristics.

DEM & Terrain Analysis

Prepared DEM-based terrain outputs including elevation, slope, aspect, hillshade, drainage and watershed-related analysis.

Gandak River Water-Quality GIS Research

Applied GIS, water-quality indices and spatial/statistical methods to investigate water-quality variation and pollution pressures in the Gandak River, Bihar.

RESEARCH PUBLICATION

“Climate Resilient in Agriculture: Investigating the Role of Millets in Sustainable Development Goals.” International Journal of Applied Science (IJASS), ISSN 2394-1405, DOI: 10.36537/IJASS/11.98/10/416-421.

PROFESSIONAL TRAINING & CERTIFICATIONS

GIS, Mapping, and Spatial Analysis Specialization *University of Toronto | Coursera | 2023*

Four-course specialization covering Introduction to GIS Mapping, GIS Data Acquisition and Map Design, Spatial Analysis and Satellite Imagery in a GIS, and a GIS/Spatial Analysis Capstone.

Fundamentals of GIS *University of California, Davis | Coursera | 2023*

Foundational GIS concepts and applications.

Spatial Analysis and Satellite Imagery in a GIS *University of Toronto | Coursera | 2022*

Spatial analysis and satellite imagery applications in GIS.

GIS Data Acquisition and Map Design *University of Toronto | Coursera | 2022*

GIS data acquisition and map design.

Introduction to GIS Mapping *University of Toronto | Coursera | 2022*

Introductory GIS mapping concepts and practice.

PROFESSIONAL PORTFOLIO

GitHub GIS & Remote Sensing Portfolio: LULC (2010–2025), NDVI, Land Surface Temperature, DEM/Terrain Analysis, and QGIS projects.

ADDITIONAL ACADEMIC SKILLS

Academic research and writing; curriculum development; student research guidance; public speaking; online quiz and assessment design; data interpretation; strategic problem solving.